

INTEGRATION AND TESTING OF THE IEEE-GRSS OPEN POCKETQUBE KIT

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ABSTRACT

With CubeSats having evolved beyond their initial educational purpose and now being employed across a diverse range of applications, more and more private enterprises are deploying large constellations for a plethora of applications. This increased popularity and enhanced performance of the nanosatellite form factor has led to heightened demands for reliability and cost efficiency. Presently, commissioning subsystem providers has become challenging, with a prevailing trend towards seeking fully integrated platforms in the market. Paradoxically, CubeSats are now less accessible to universities and research institutions than they were a few years ago.

To address these challenges, the PocketQube concept emerged. Making up the picosatellite class, and measuring 50x50x50 mm³, they provide a cost-effective alternative, particularly for educational purposes. Students working with PocketQubes encounter design challenges akin to those faced with larger satellites. This paper comes in continuation of the IEEE-GRSS "Open PocketQube Kit" educational initiative, developed by the NanoSat-Lab at the Universitat Politècnica de Catalunya and is focused along the integration and testing of the subsystems, as well as performing a full system validation and an environmental testing.

The initiative has resulted in the development of three distinct PocketQube models: ^{Po}Cat-1 equipped with a VGA camera, and ^{Po}Cat-2 and ^{Po}Cat-3 designed for monitoring radio-frequency interference at L-band (1-2 GHz) and K-band (24-25 GHz) respectively. The latter two have undergone a series of drone campaigns in the first half of 2024, sponsored by the Frequency Allocations in Remote Sensing (FARS) Technical Committee of the IEEE Geoscience and Remote Sensing Society (GRSS).

Index Terms— SmallSats, PocketQubes, Education, New-Space, Earth Observation, Integration, Testing.



Fig. 1: PocketQube 1 with folded COMMS antenna

1. INTRODUCTION

As previously mentioned, the primary objective of the IEEE-GRSS Open PocketQube kit [1] for New-Space missions is its educational component, particularly targeting newly established institutions and organizations seeking entry into the space sector. Over the past 8 semesters, this kit has undergone significant development, spearheaded by undergraduate students from the School of Telecommunications Engineering at the Universitat Politècnica de Catalunya (UPC) within the NanoSat Lab [2] with over 70 persons working in the project, which has now reached its final leg of the journey, with only the final validations left to complete.

Additionally, the kit has been accepted and is currently participating in the European Space Agency's Fly Your Satellite 4 [3] programme, with the main goal of further refining the testing procedures and to participate in the subsequent provided launch.

2. SUBSYSTEM VERIFICATION

Each of the picosatellite's subsystems carries out very specific and crucial tasks, briefly explained as follows:

- On Board Computer (OBC) is a critical component of a satellite, comprising a basic microcontroller. It plays a key role in overseeing and directing various satellite processes through a multiple thread architecture. Additionally, the OBC is responsible for making critical decisions based on diverse situations that may arise within the system.
- Communications System (COMMS) based on the LoRaWAN protocol [4]. A Federation Deployment Control Protocol (FeDeCoP) [5] has been implemented in order to allow the PoCat trio to act like a satellite federation, exchanging data constantly and dynamically allocating unused resources such as memory, power or downlink opportunities, taking advantage of their innate low cost and relatively easy mass production to improve the otherwise limited capabilities of the PocketQubes.
- Electrical Power Supply (EPS) harvests the solar energy obtained using an array of triple junction Gallium-Arsenide Solar cells, manages the charging and discharging of the lithium polymer battery and regulates the power flow, all the while allowing the OBC full control over all of these processes.
- Attitude Determination and Control System (ADCS) performs the complex tasks of both assessing the relative orientation of the spacecraft using an array of magnetic field sensors and photodiodes, as well as adjusting the attitude using a set of magnetorquer actuators needed for when any of the satellite's routines require pointing.
- A Remote Sensing Payload, which at the time of writing this article comes in one of the following variants: VGA camera, a L-band RFI monitor designed to determine radio pollution threatening both the geolocation services as well as the protected radiometer band present in the 1 to 2 GHz band [6], and lastly a K-band RFI monitor with a similar objective, concerning the water vapour monitoring band (24-25 GHz), where some 5G services have been allocated.

As previously mentioned, all these subsystems require specific and extensive testing, with detailed procedures redacted but, for brevity's sake, not described in the present manuscript.

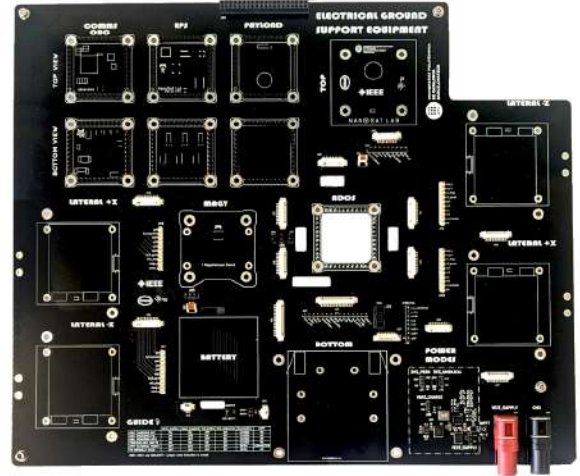


Fig. 2: Electrical Ground Support Equipment

3. GROUND SUPPORT EQUIPMENT

In order to facilitate system testing, the Electrical Ground Support Equipment (EGSE) seen in Fig. 2 has been developed, following a flat-sat ideology, allowing for seamless connection of all subsystems with a multitude of marked switches and connectors allowing validation of all operation modes of the satellite, with the added benefit of not requiring a full integration procedure for each test conducted. The EGSE comes itself with a fully functional EPS, identical to the one used on board the satellite, thus permitting greater fidelity testing using any combination of inputs, such as solar array power, battery or external supply and varying degrees of satellite power consumption also being controllable depending on the operating mode.

Moreover, another benefit of the flat-sat is that, with all boards inserted into the multiple sockets over the testbed's surface, debugging becomes easier, as everything is readily accessible, as opposed to the post-integration testing, where the inner boards become practically untouchable from a testing standpoint.

Additionally, a test pod has been created to assess the satellite's response to high vibration levels during launch. This evaluation involves the use of an Electrodynamics Vibration Table for quasistatic, random, and sine vibration analysis. The pod comprises an IC306 stainless steel adaptation plate and a fastener assembly made from Aluminum 6063-T6 and can be seen in Fig. 3, with both PocketQube variants 2 and 3 with their antennas deployed. It was designed to closely adhere to the PocketQube standard, ensuring an accurate replication of both the shape and mechanical characteristics of the deployer. Notably, this design allows for the simultaneous loading and testing of two PocketQubes in direct contact, better simulating the conditions of an actual deployment.

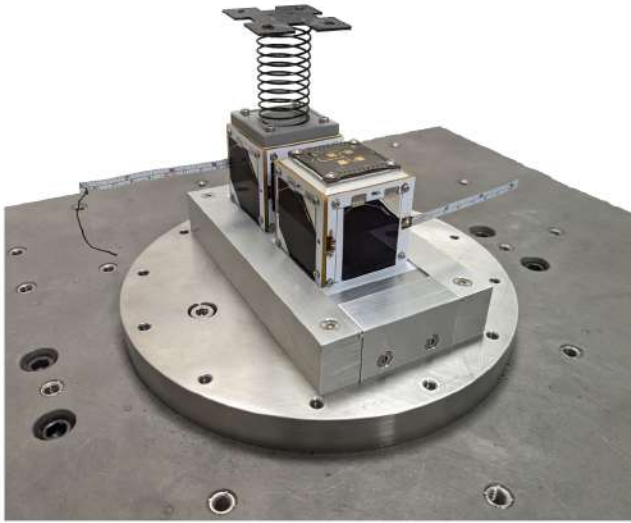


Fig. 3: Mechanical Ground Support Equipment with K-band (front) and L-band (brack) PocketQubes integrated

4. FULL SATELLITE INTEGRATION AND ENVIRONMENTAL TESTING

After both Subsystem Verification and EGSE validations are passed, the next step would be to fully ensemble the picosatellite. Fig. 4 shows the inside view of the PocketQube. Starting from the bottom, there is the lithium polymer battery, inside its designated PTFE housing, with the material chosen for its good thermal insulation properties, as well as compliance with the envelope constraints imposed by [7]. Next up, the avionics of the satellite are stacked vertically and interconnected using a standard, board to board set of SMD connectors and fastened to the bottom and top structure pieces using an ensemble of screws and spacers for improved stiffness. Lastly, the lateral boards are connected using wire to board connectors to the ADCS board, which doubles as a break-out point for the star configuration connectivity scheme of the outer elements.

Next, a minimal set of functional tests are to be carried out, qualifying the picosatellite before the subsequent GEVS-compliant [8] Thermal Vacuum Chamber and Shake Table

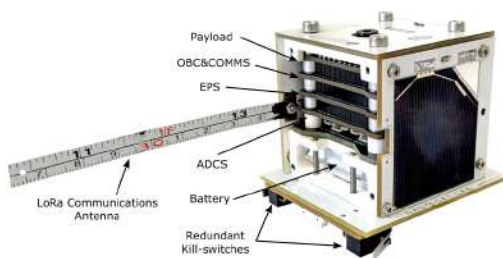


Fig. 4: Open PocketQube with visible inner boards

campaigns, with another round of functional tests to be performed after each campaign, thus validating the spacecraft's correct operation under and after extreme conditions.

5. CONCLUSIONS

This paper aims at describing the testing procedures for a set of educational PocketQubes with highlights on the environmental tests and assembly procedures. All the findings will be presented in greater detail at the conference.

6. ACKNOWLEDGMENTS

This work has been developed as part of the IEEE Geoscience and Remote Sensing Society "Open PocketQube Kit" educational initiative and it will be openly distributed when completed.

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